

Virtual ASR33 Teletype

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Running

There are example “bash” scripts showing how to run the telnet and tty (serial port) variants. Serial ports require the external package jSerialComm, which is a submodule in git repository. The general commandline syntax is:

Telnet version

```
java -jar ASR33telnet.jar [prop=value...] [host port]
```

Tty version

```
java -cp ${JARS} ASR33tty [prop=value...] [tty baud]
```

Where “\${JARS}” here represents the paths to the jar files required to run, specifically ASR33tty.jar and jSerialComm-2.9.3.jar, separated by a colon (Linux/MAC) or semi-colon (Windows).

Controls

The menu item **PTape ► Punch** selects a file to use for the paper tape punch. Any previous content is lost. Clicking "Cancel" leaves no file selected (punch not operational). Editing (backspacing, the B.SP. button) is not supported.

The menu item **PTape ► Reader** selects a file to use for the paper tape reader. In order to re-wind a file, re-select it from this menu. Clicking "Cancel" leaves no file selected (reader not operational).

LOCAL checkbox selects between LOCAL (checked) or LINE (unchecked) mode of operation. In LOCAL mode all characters from the remote are dropped and keys pressed are not sent to the remote but rather printed locally.

If a file has been mounted on the Punch device, the PUNCH checkbox indicates whether the punch is on - in which case anything printed will also be punched.

If a file has been mounted on the Reader device, the READER checkbox indicates whether the reader is enabled (ready – AUTO on some models) - in which case it is turned on and off by X-ON and X-OFF characters or does a single-character read if a command is received.

This implementation recognizes the SI character (Ctrl-O, 0x0F) to trigger a single character read from tape. That character may be changed with the "rdr_adv_char" property. Only control characters are tested for this character, so setting to any value greater than or equal to 32 (0x20, SP) effectively disables the feature.

If the Reader is in the ready state, the "start" button may be used to start reading tape and sending to the remote. Whenever the reader is active, the "start" button will be illuminated. Pressing this button while the reader is active will stop the reader.

The menu item **PTape ► Rdr_Position** will pop-up a window allowing positioning of the paper tape in the reader. The reader is disabled during this operation (same as being in the FREE position). The keys UP, DOWN, **Page_Up**, and **Page_Down**, along with the mouse wheel, will reposition the tape. **Home** jumps to the beginning, and **End** jumps to the end. Note that the beginning of the tape is “down” in the view, and the keys/actions are moving the *tape*, not the view. Press **Enter** or Close the window to set the new position and release the reader. Press **Delete** to cancel without changing position.

The menu item **PTape ► Pun_Viewer** will pop-up a window allowing viewing of the paper tape in the punch. The keys UP, DOWN, **Page_Up**, and **Page_Down**, along with the mouse wheel, will reposition the tape. **Home** jumps to the beginning, and **End** jumps to the end. Note that the beginning of the tape is “down” in the view, and the keys/actions are moving the *tape*, not the view. Press **Enter**, **Delete** or Close the window. The position of tape in the punch is never changed.

The menu item **Print ► Tear_Off** will discard all text currently “printed”, clearing the window.

The menu item **Print ► Save** allows saving all text currently “printed” to a file.

A scroll bar on the right side allows looking back in currently “printed” text. There is currently no way to view/monitor paper tape being punched, nor is there a “live view” of the paper tape reader.

Keyboard

F12 functions as the HERE IS key. The answer-back drum defaults to 20 NUL characters (often used to make blank leader/trailer on punched paper tape), but may be set to a string using the "ansbak" property (20 characters maximum).

The Enter key produces CR, use Ctrl-J or Ctrl-Enter for LF.

All characters in the last ASCII-7 bank (lower case letters, etc) except DEL (0x7f) are folded into the upper-case bank.

The Delete key produces the RUBOUT character 0x7f.

The menu item **Print ► Copy** (also Alt-C) will copy any selected text (the usual mouse click-drag) to the clipboard.

The menu item **Print ► Paste** (also Alt-V or middle mouse button) will paste text from the clipboard as keyboard input. This does not paste text to the “printed page”.

Misc

Two modes are possible for the telnet version, "server" or client (default).

Characters transmitted for the Reader or HERE IS function, or pasted into the window, are delayed according to the "delay" property, the number of milliseconds to delay after each character. Default is 100 milliseconds. The CR may incur a different delay according to the “cr_delay” property. Default CR delay is 1000 milliseconds.

Pasted text being transmitted may be canceled by pressing the F1 key.

Telnet version

(client) If the connection to the remote is lost or not present to begin with, the menu item **Print ► Reconnect** may be used to attempt to re-connect to the remote using the same host and port.

(client) A new, different, connection may be specified using the **Print ► New_Connection** menu item.

The current connection may be dropped using the **Print ► Hangup** menu item.

Tty version

The menu item **Screen ► New_Connection** will prompt for a new tty and baud and then attempt to close and re-open the serial port.

Properties

Properties with the "asr33_" prefix may be specified on the commandline by removing the prefix. Quotes around values may be required, depending on your shell. Boolean properties do not require a value, but do require a “=” when used on the commandline.

The terminal will search for properties in the following order:

1. *prop=value* arguments on the commandline
2. Using the environment variable ASR33_CONFIG
3. Looking for "asr33.rc" in the current directory
4. Looking for ".asr33rc" in the user's home directory

Only one properties file will be used - they are not combined. Commandline properties will override (or supplement) whichever properties file was loaded.

asr33_ansbak=*string*

Specifies a string to use for the Answer-back drum. Typically, this drum generates 20 NUL characters, and that is the default. A maximum of 20 characters is used, with the remaining positions on the drum being left as disabled. This string may include the standard "escapes" for special characters, such as "\r" for CR and "\n" for LF.

asr33_delay=ms

The delay inserted after each character is pasted or otherwise automatically sent to the host, in milliseconds. Default is 100. This does not affect typing from the keyboard.

asr33_cr_delay=ms

The delay inserted after each CR is pasted or otherwise automatically sent to the host, in milliseconds. Default is 1000. This does not affect typing CR from the keyboard.

asr33_rdr_adv_char=value

tty_rdr_adv_char=value

The control character to use for performing a single-character read operation ("advance") on the paper tape reader. This emulates the Teletype feature of a solenoid on the reader that is activated by a handshake signal on the host interface. This needs to match whatever is used on the host (e.g. MDS-800) to trigger the reader advance. Default is 0x0f (Ctrl-O). Setting to a non-control character value (>=32) effectively disables the feature.

asr33_rdr_view=num

The number of characters before and after the current in the reader tape positioning view. Default is 8. Must be between 1 and 30.

asr33_bell=wav-file

The audio file to use for the bell sound. This file is buffered in memory, so needs to be small. Default is "bell.wav".

asr33_bell_volume=percent

Specifies the audio volume to use for the bell, in percent. Default is "50".

asr33_lines=num

Specifies the number of lines to view in the text window. Minimum is 10. Default is 24. Scrolling still allows viewing the entire printed text, this property only affects the view size. Beware of impractical numbers for your screen size.

asr33_font_size=points

Specifies the size of the font (in points) to use. Default is 12. Minimum is 4 and maximum is 30.

asr33_live_ppt

Enables a "live view" of the paper tape reader and punch. The views are fixed windows showing only the most-recently punched data or reader data before/after the next character to read.

Teletype Option Properties

The following properties accept the values (case-insensitive) "yes", "true", "on", or "1" to enable an option. All other values disable. If the property is not set, the default value is used.

asr33_auto_nl

Enables the automatic CR-LF at column 80. Default is on.

asr33_auto_rdr

Enables the paper tape reader to respond to DC1/DC3. Default is on.

asr33_auto_pun

Enables the paper tape punch to respond to DC2/DC4. Default is on.

asr33_cr_on_if

Enables the automatic CR when LF is received. Default is off.

asr33_if_on_cr

Enables the automatic LF when CR is received. Mutually exclusive with **asr33_cr_on_if**. Default is off.

asr33_nonprint

Enables the Print/Nonprint option, which adds a button to control whether or not received characters are printed (they are always punched if the punch is on). Default is off.

asr33_parity

Enables generation of parity in the 8th bit of keyboard codes. Default is off.

asr33_even

Selects the state of the 8th bit of keyboard codes. Default is on when parity is enabled (EVEN parity), off when parity disabled (SPACE parity bit).

asr33_col72_bell

Used to disable the margin bell after column 72. Default is enabled.

Telnet Properties

asr33_host=host

The hostname or IP address to connect to. This is the same as the first non-property argument on the command line.

asr33_port=port

The port number to connect to. This is the same as the second non-property argument on the command line.

asr33_server

Sets server mode, where the ASR33 listens on the specified host and port, simulating a standalone case where the ASR33 waits for connections from remote ASR33s. Value is ignored, but must be present ("=" at least) when given on the commandline. Property must be absent to turn off.

Tty Properties

asr33_tty=device

The device path of the serial port to connect to. This is the same as the first non-property argument on the command line.

asr33_baud=speed

The speed (baud) to use on the selected serial port. This is the same as the second non-property argument on the command line.